

Kelly criterion proofs

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Introduction

Theorem

1. If $G(f) > 0$, then, $\lim_{n \rightarrow \infty} X_n = \infty$ a.s., that is, for each $M > 0$,
 $P(\liminf_{n \rightarrow \infty} X_n > M) = 1$.
2. If $G(f) < 0$, then, $\lim_{n \rightarrow \infty} X_n = 0$ a.s., that is, for each $\epsilon > 0$,
 $P(\limsup_{n \rightarrow \infty} X_n < \epsilon) = 1$.
3. If $G(f) = 0$, then $\limsup_{n \rightarrow \infty} X_n = \infty$ a.s., and, $\liminf_{n \rightarrow \infty} X_n = 0$ a.s.
4. If $G(f_1) > G(f_2)$, then, $\lim_{n \rightarrow \infty} \frac{X_n(f_1)}{X_n(f_2)} = \infty$ a.s.

Proof.

(1) **Claim:** $P\left(\lim_{n \rightarrow \infty} \log \left(\frac{X_n}{X_0}\right)^{1/n} = G(f)\right) = 1.$

Suppose the claim is true. Then, since $G(f) > 0$, using the claim it follows that a.s. for $\omega \in \Omega$, there exists $N(\omega)$ such that for all $n \geq N(\omega)$,

$$\log \left(\frac{X_n}{X_0}\right)^{1/n} \geq \frac{G(f)}{2} > 0,$$

which further implies

$$X_n \geq X_0 e^{\frac{n}{2} G(f)} \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

Hence, $X_n \rightarrow \infty$ as $n \rightarrow \infty$ a.s.

Now, to prove the claim, note that S_n and F_n are binomial random variables, hence by Borel's Strong Law of Large numbers it follows that

$$\lim_{n \rightarrow \infty} \frac{S_n}{n} = p, \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{F_n}{n} = q$$

with probability 1, that is,

$$\lim_{n \rightarrow \infty} \frac{S_n}{n} \log(1+f) + \lim_{n \rightarrow \infty} \frac{F_n}{n} \log(1-f) = p \log(1+f) + q \log(1-f),$$

with probability 1, whence,

$$\lim_{n \rightarrow \infty} \log \left(\frac{X_n}{X_0} \right)^{1/n} = G(f),$$

with probability 1. □

(3) We will use the fact that given any M , $\overline{\lim}_n [S_n - np] \geq M$. Then if, $S_n \geq np + M$, from eq.(??), we obtain

$$\log \left(\frac{X_n}{X_0} \right)^{1/n} \geq \frac{np + M}{n} \log(1+f) + \frac{n - (np + M)}{n} \log(1-f),$$

which implies

$$\log \left(\frac{X_n}{X_0} \right)^{1/n} \geq p \log(1+f) + q \log(1-f) + \frac{M}{n} [\log(1+f) - \log(1-f)]$$

$$= G(f) + \frac{M}{n} \log \left(\frac{1+f}{1-f} \right) = \frac{M}{n} \log \left(\frac{1+f}{1-f} \right).$$

Now, as $G(f) = 0$,

$$\frac{X_n}{X_0} \geq \left(\frac{1+f}{1-f} \right)^M.$$

Since, $S_n \geq np + M$ infinitely often a.s., it follows that

$$\overline{\lim} X_n \geq X_0 \left(\frac{1+f}{1-f} \right)^M,$$

and hence, as $M \rightarrow \infty$, $\overline{\lim} X_n = \infty$ a.s. □

(4) Again, from eq. (??), for the fractions f_1 and f_2 , we have

$$\log \left(\frac{X_n(f_1)}{X_0} \right)^{1/n} - \log \left(\frac{X_n(f_2)}{X_0} \right)^{1/n} = \frac{S_n}{n} \log \left(\frac{1+f_1}{1+f_2} \right) + \frac{F_n}{n} \log \left(\frac{1-f_1}{1-f_2} \right),$$

that is,

$$\log \left(\frac{X_n(f_1)}{X_n(f_2)} \right)^{1/n} = \frac{S_n}{n} \log \left(\frac{1+f_1}{1+f_2} \right) + \frac{F_n}{n} \log \left(\frac{1-f_1}{1-f_2} \right).$$

Applying Borel's Strong Law of Large Numbers,

$$\frac{S_n}{n} \log \left(\frac{1+f_1}{1+f_2} \right) + \frac{F_n}{n} \log \left(\frac{1-f_1}{1-f_2} \right) \rightarrow G(f_1) - G(f_2),$$

with probability 1, as $n \rightarrow \infty$, that is,

$$\log \left(\frac{X_n(f_1)}{X_n(f_2)} \right)^{1/n} \rightarrow G(f_1) - G(f_2),$$

with probability 1, as $n \rightarrow \infty$. But as $G(f_1) - G(f_2) > 0$, using part (1), it follows that

$$\lim_{n \rightarrow \infty} \frac{X_n(f_1)}{X_n(f_2)} = \infty \text{ a.s.}$$

□

Theorem

If for each i , there is an f_i , $0 < f_i < 1$ such that $E[\log(1 + F_i Q_i)]$ is defined and positive, then for each i , there is a number f_i^ such that $E[\log(1 + F_i Q_i)]$ attains its unique maximum for $F_i = f_i^*$ a.s. (To avoid trivialities we assume $Q_i \neq 0$ for each i .)*

Proof. The domain of definition of $E[\log(1 + F_i Q_i)]$ is an interval $[0, a_i]$ or $[0, a_i)$, where $a_i = \min\{1, b_i\}$, and, $b_i = \sup\{f_i : f_i Q_i > 0 \text{ a.s.}\} > 0$. Since, the second derivative of $E[\log(1 + f_i Q_i)]$ is $-E[Q_i^2/(1 + f_i Q_i)^2]$, which is defined and negative, any maximum of this function is unique. Also, the function $E[\log(1 + F_i Q_i)]$ always has a maximum even if it is not defined on a_i , as $\lim_{f_i \rightarrow a_i} E[\log(1 + F_i Q_i)] = -\infty$. Now,

$$E[\log(1 + F_i Q_i)] \leq E[\log(1 + f_i^* Q_i)],$$

where the equality holds iff $E[\log(1 + F_i Q_i)] = E[\log(1 + f_i^* Q_i)]$ a.s., which is equivalent to $F_i Q_i = f_i^* Q_i$ a.s., and by the independence of F_i and Q_i it follows that $F_i = f_i^*$ a.s. □